Learning about the inductive potential of categories from language

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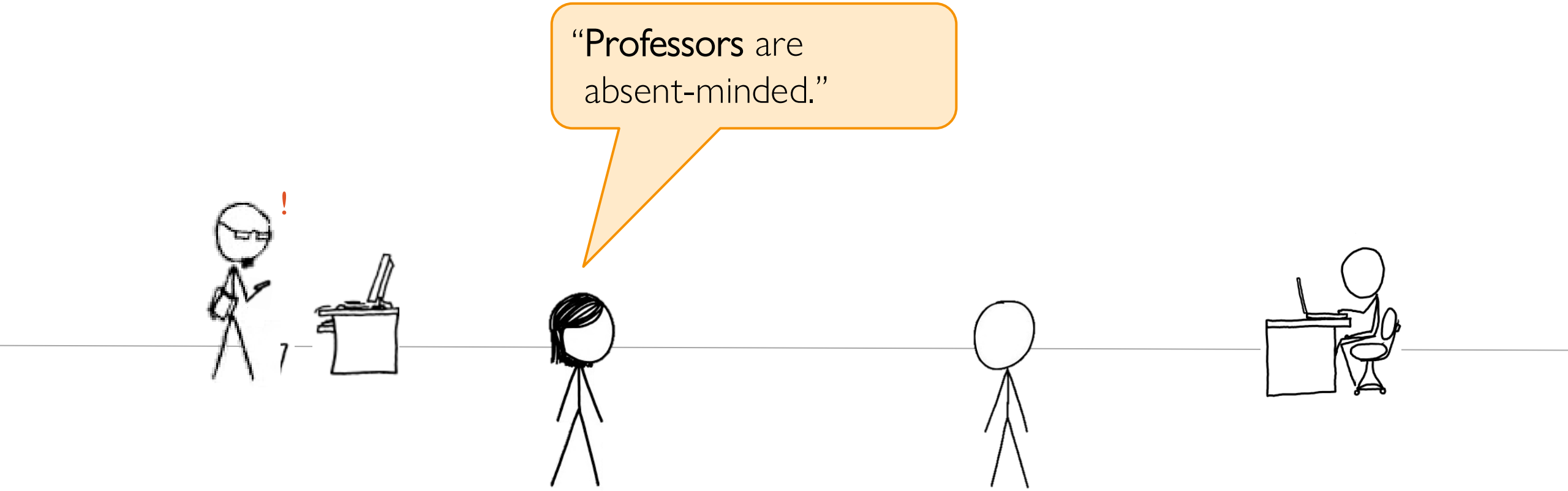
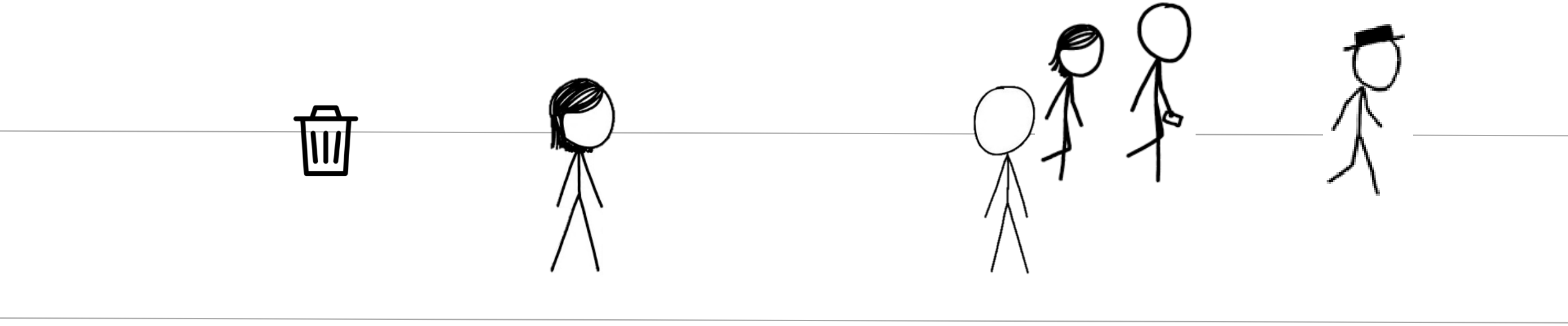
Marjorie Rhodes
Mark Ho

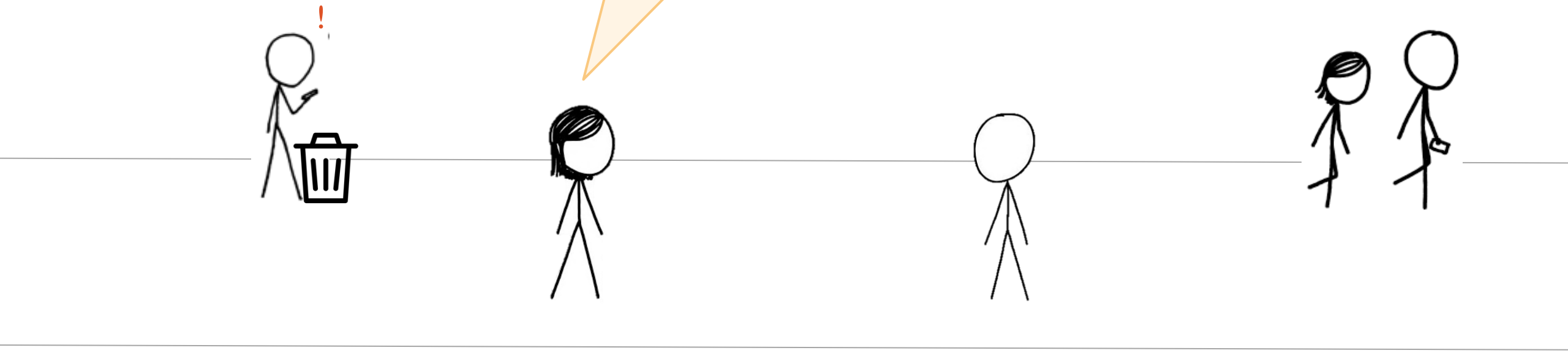
generic statement



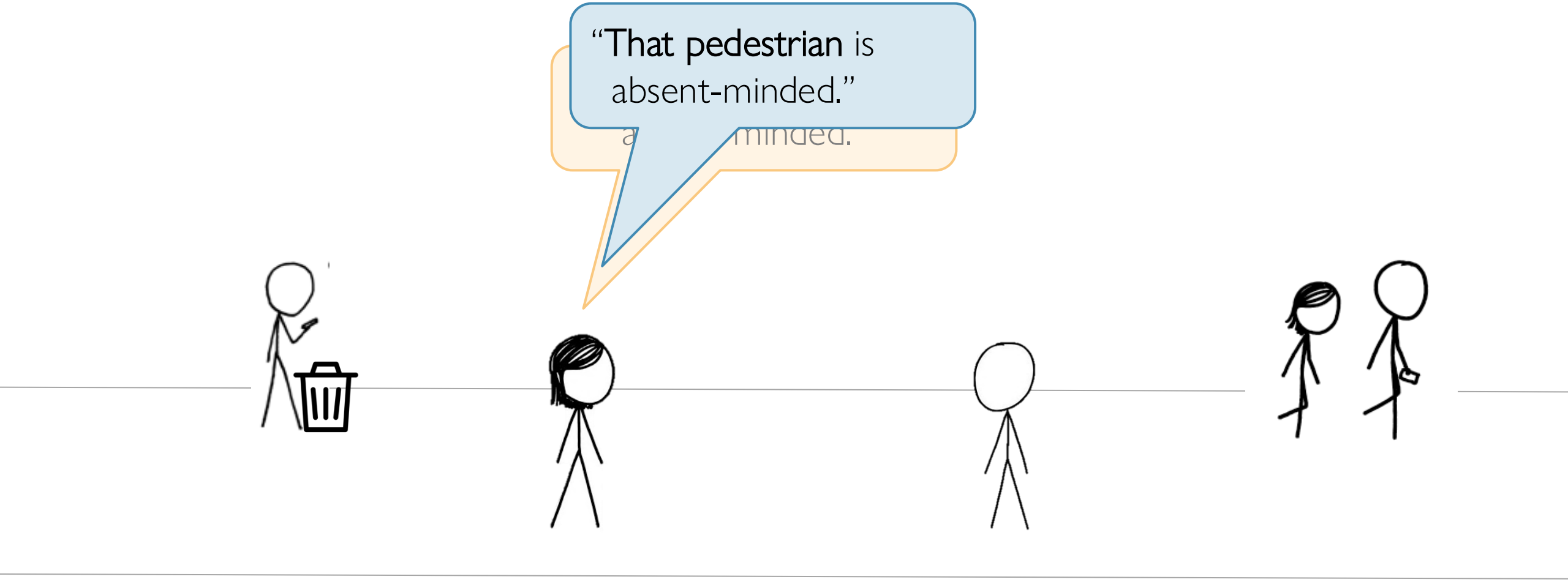


generic statement

"Pedestrians are absent-minded."



specific statement



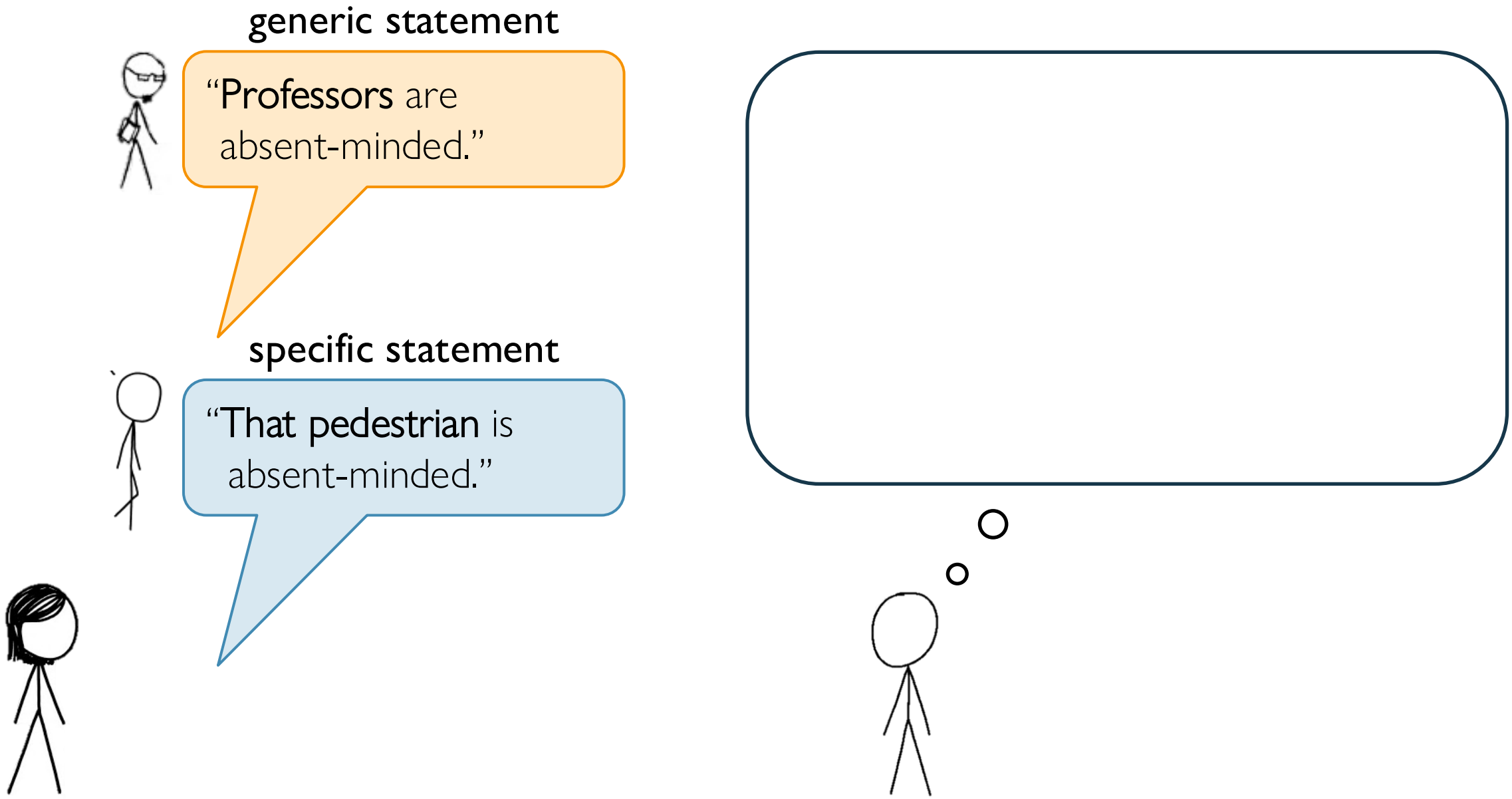
generic statement

"Professors are absent-minded."

specific statement

"That pedestrian is absent-minded."

Language can signal what social categories support generalization.


generic statement
"Professors are absent-minded."
specific statement
"That pedestrian is absent-minded."

Language as a signal of inductive potential

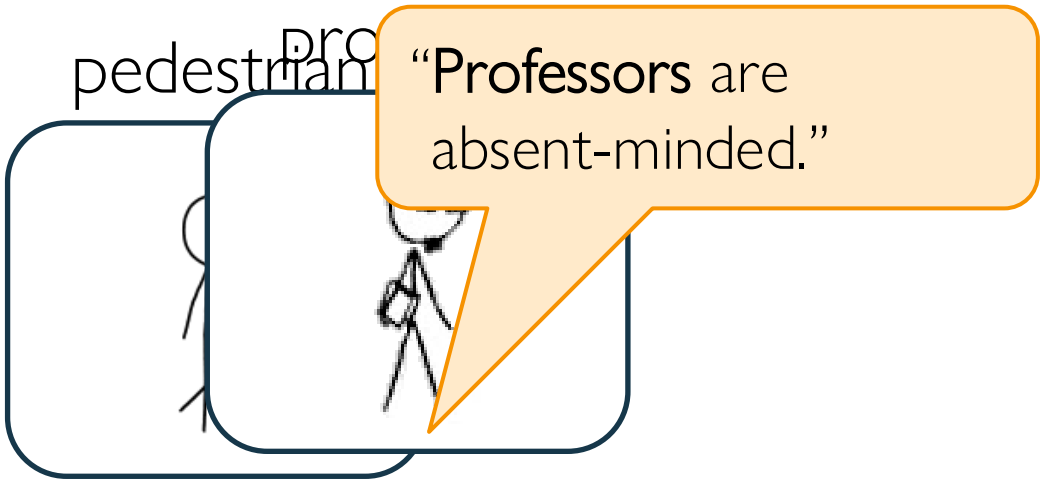
how well a category
supports generalization

low — high

categories seen as minimal,
not supportive of generalization

categories seen as coherent/meaningful,
supportive of generalization

generic statement



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"That pedestrian is absent-minded."

generic statement

"Professors are absent-minded."

professor

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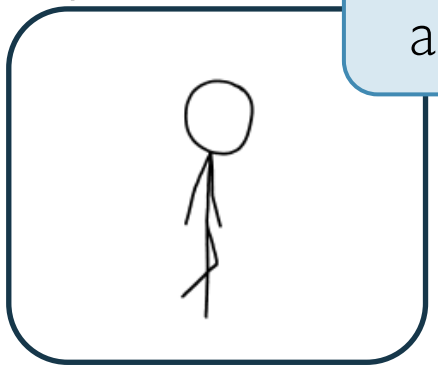
generic statement

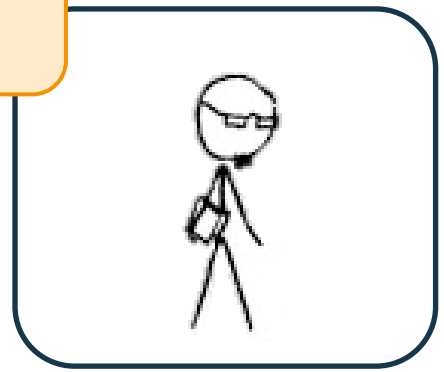
pedestrian

"**That pedestrian** is absent-minded."

"**Professors** are absent-minded."

professor

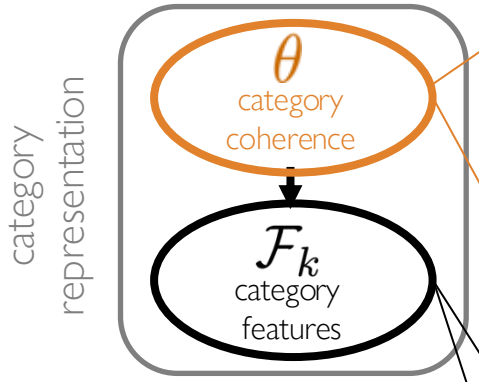




Computational model for learning about categories from language

Computational model for learning about categories from language

Learning a category representation
via hierarchical Bayesian reasoning
(Kemp et al., 2007; A. Gelman, 1995)



category coherence

an overhypothesis on the probability that any feature of an individual category member is also a category feature

category features

set of features linked to the category
e.g., {"smart", "eccentric", …} for professor

Computational model for learning about categories from language

Learning a category representation
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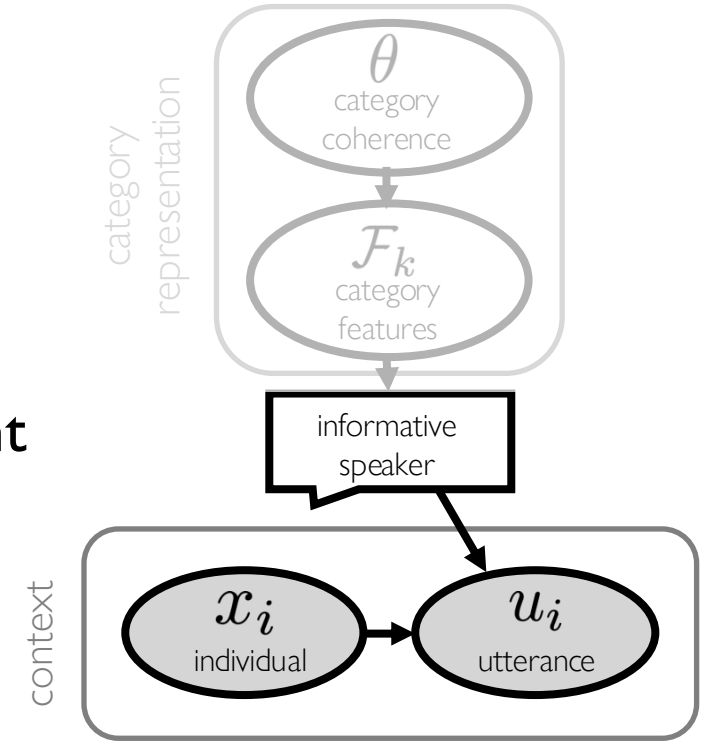
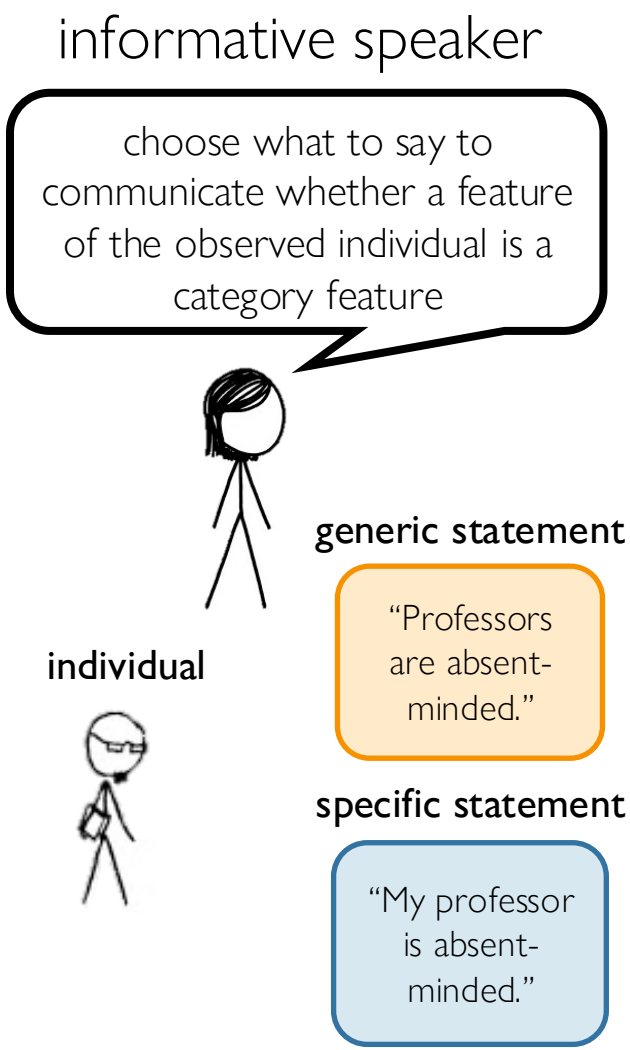
(Kemp et al., 2007; A. Gelman, 1995)

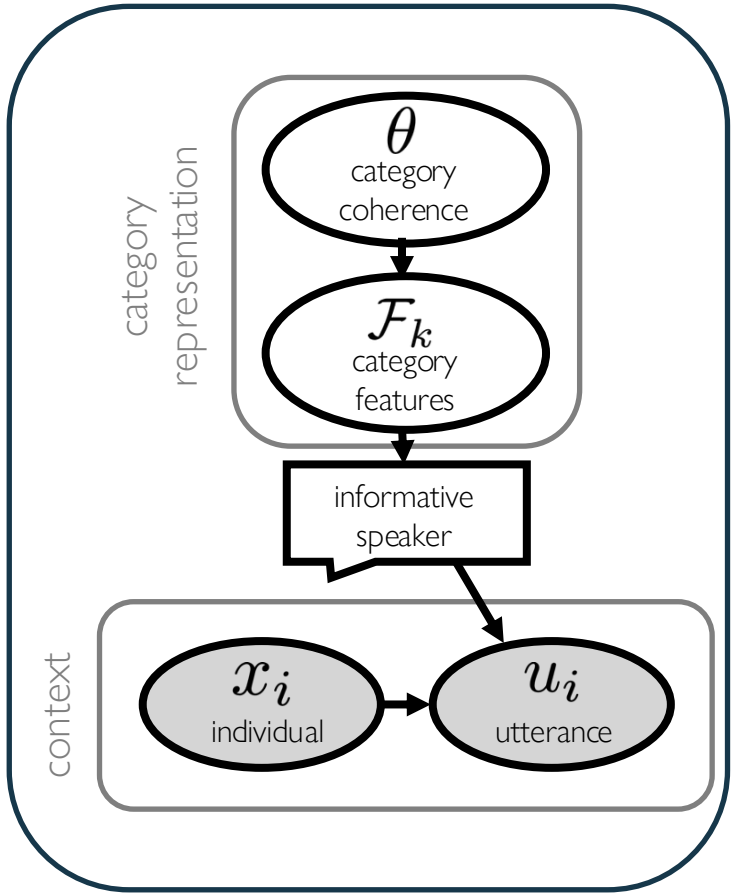
Reasoning about what a speaker meant
via rational speech acts framework
(Goodman & Frank, 2016)





Computational model for learning about categories from language

Our hierarchical Bayesian + rational speech acts model predicts that:



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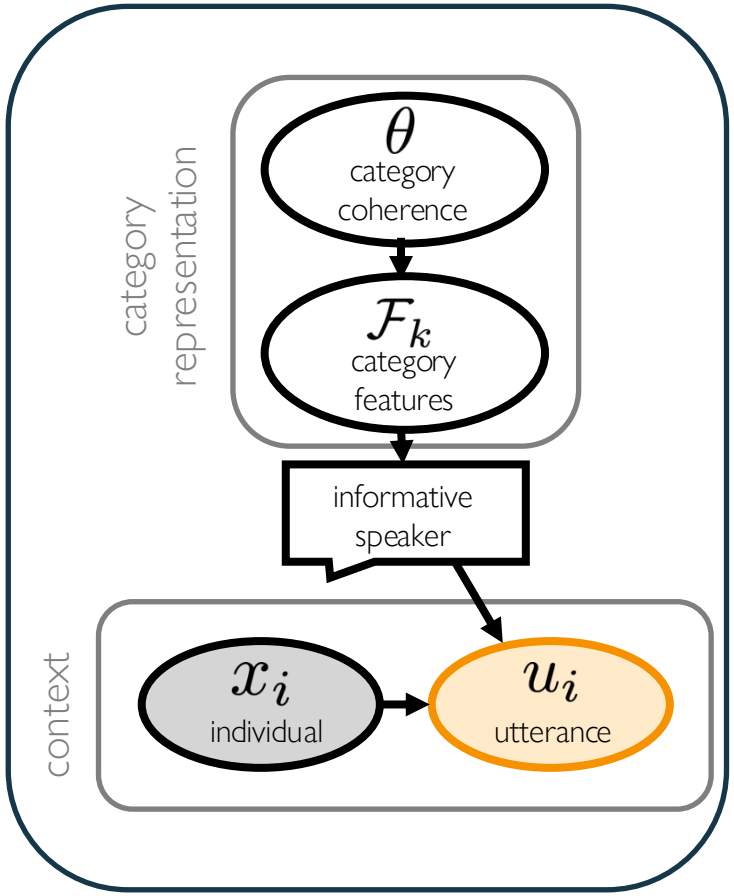

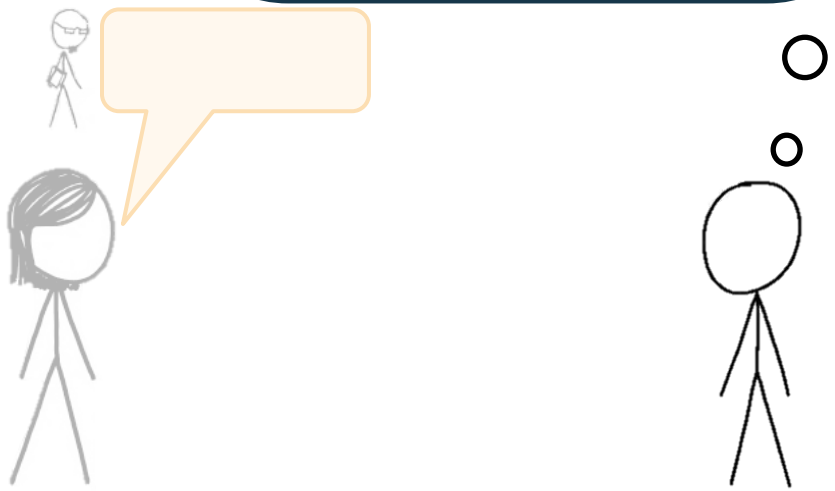

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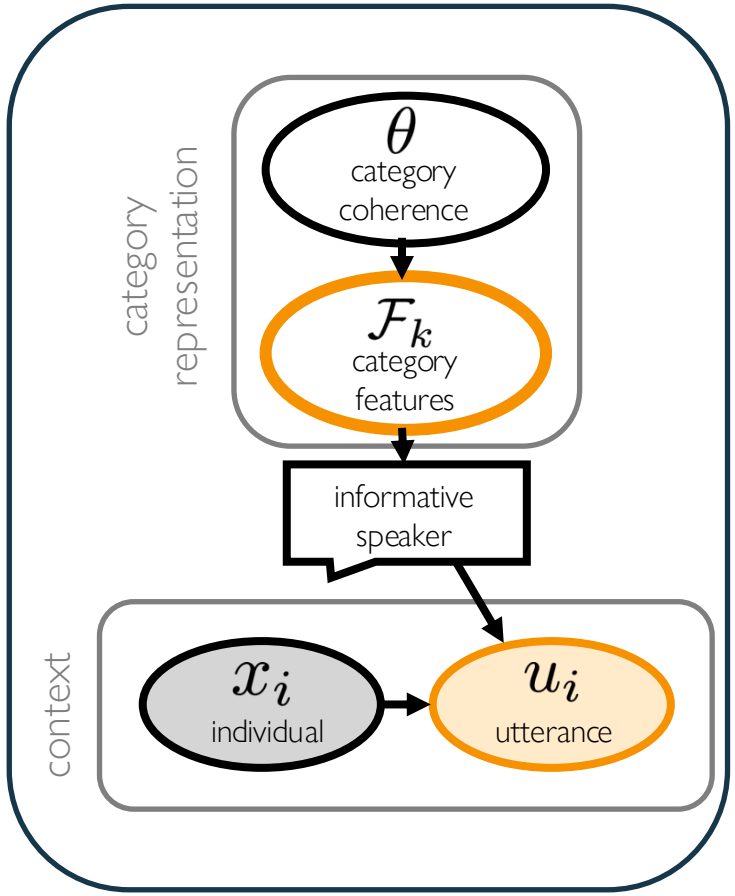

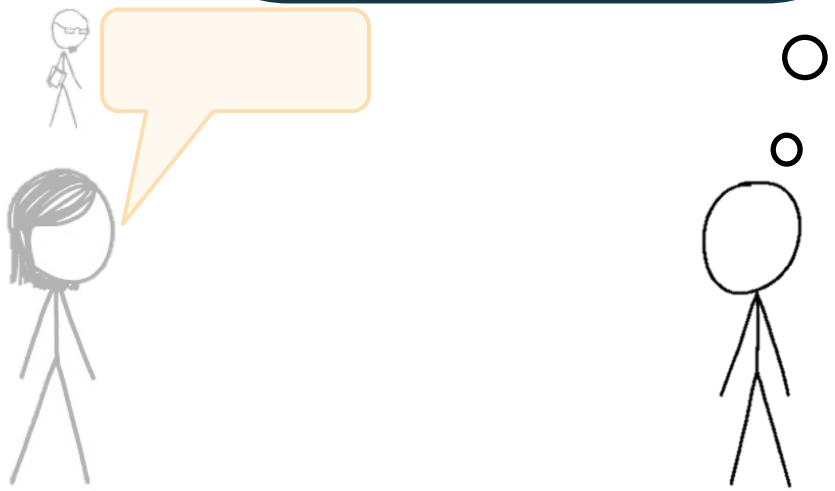

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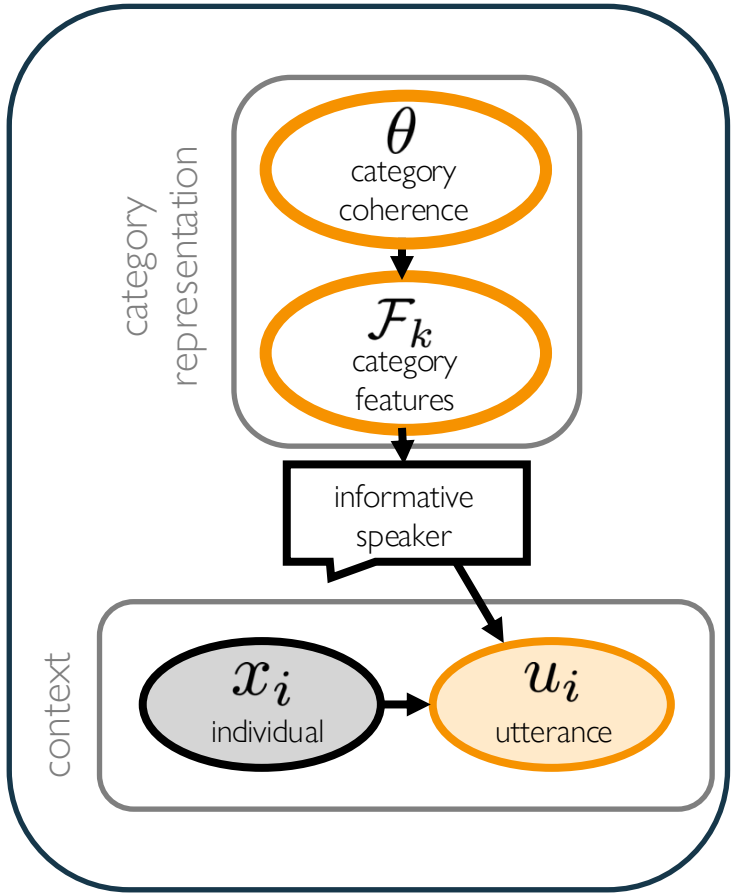

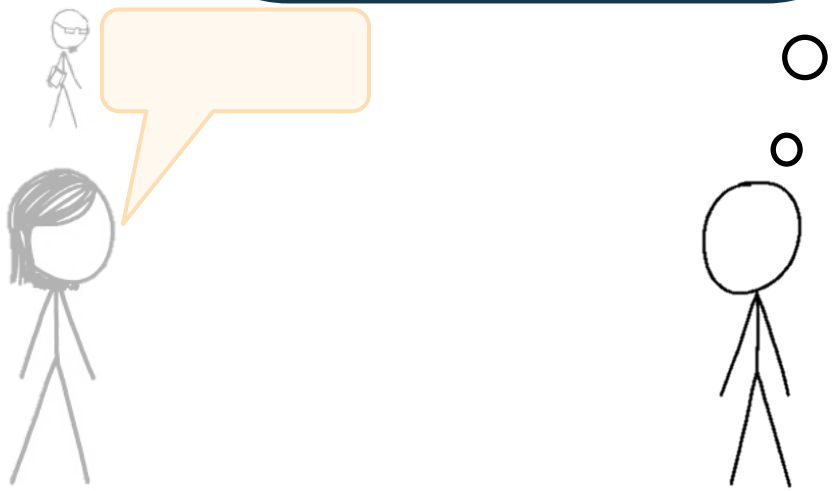

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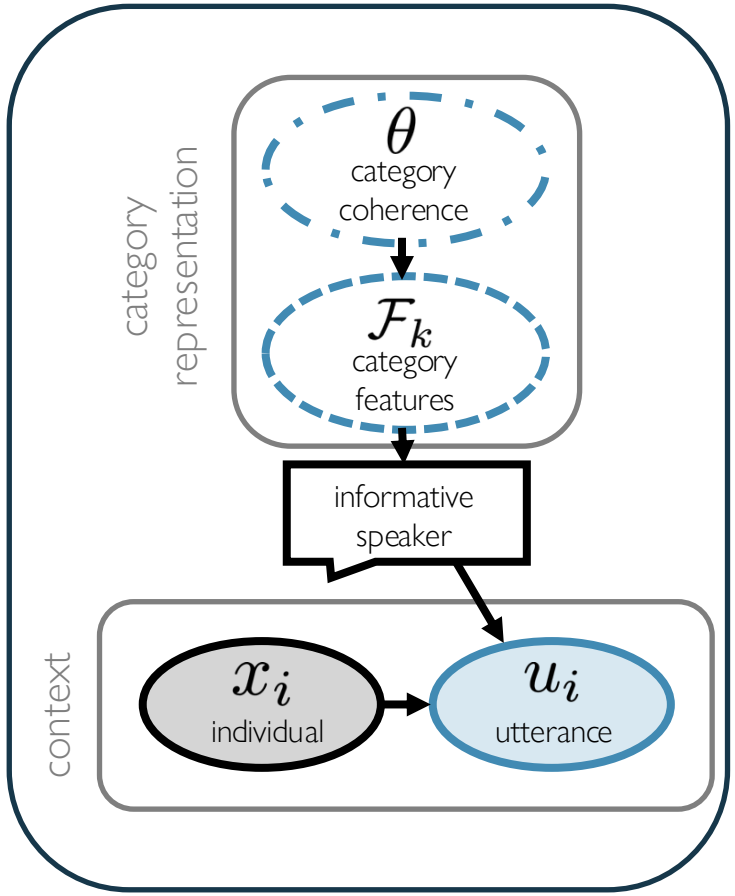
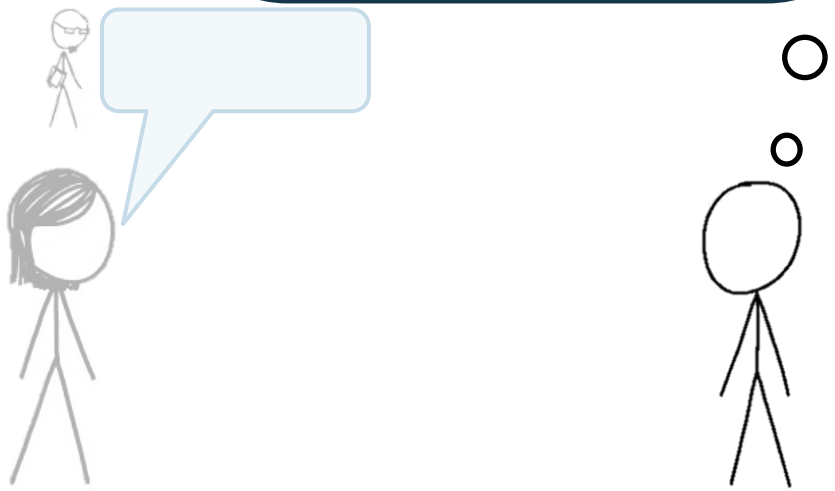
1. Hearing generic statements about a category (e.g., "Professors are absent-minded.") will increase generalization across the category.

2. Hearing specific statements about individuals (e.g., "That pedestrian is absent-minded.") will decrease generalization across the category.





Experimental design

adults learned about a novel social group, Zarpies

(US, Prolific, n=284, n=90-99/condition)

generic condition

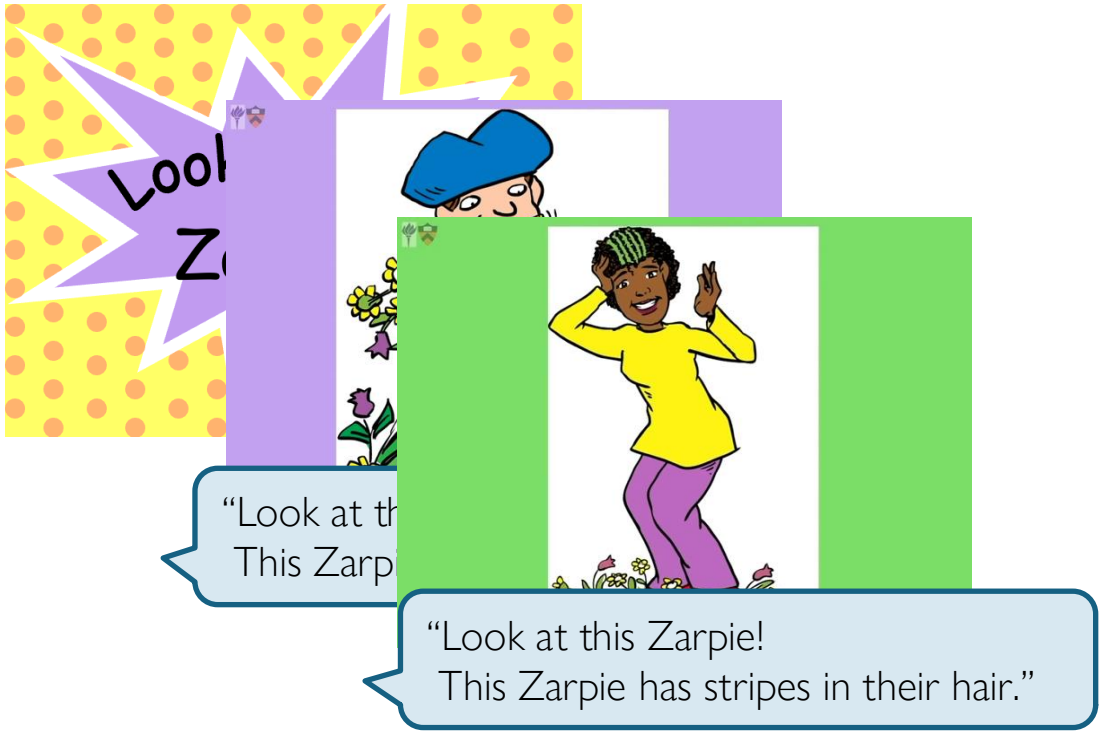
16 generics about Zarpies

baseline condition

no info

specific condition

16 specifics about individual Zarpies



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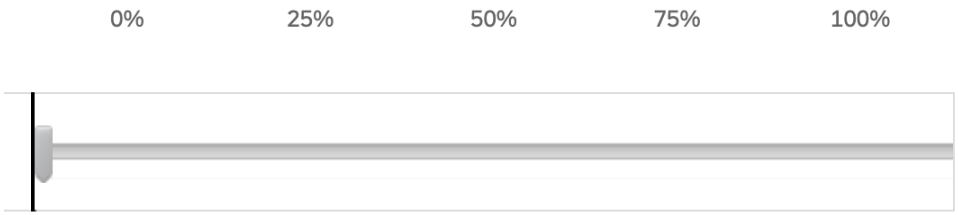
16 specifics about individual Zarpies

…then completed an inductive potential task

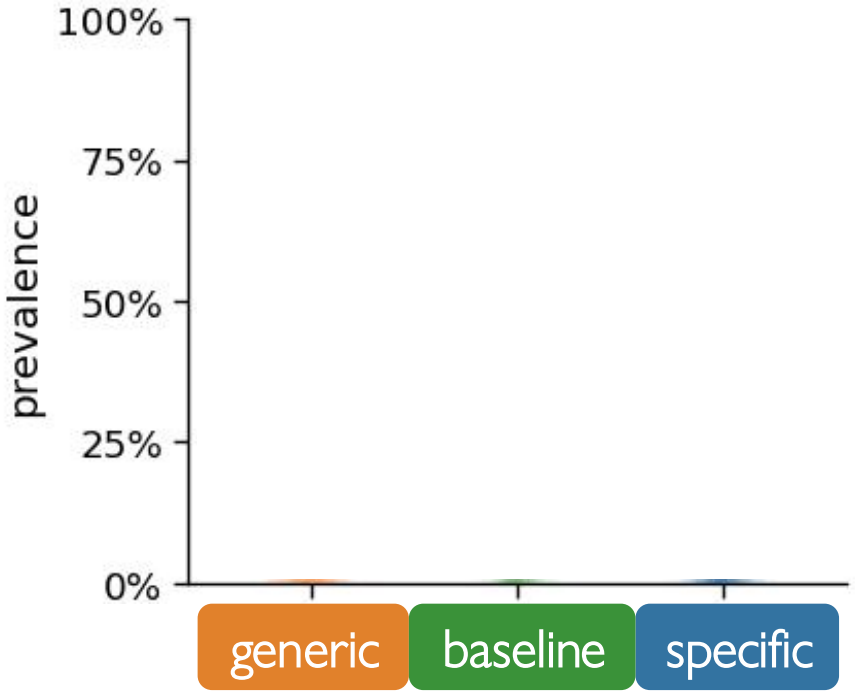
"Imagine you see a Zarpie painting their hands yellow.
What perce

"Imagine you see a Zarpie farming potatoes.
What percentage of Zarpies do you think farm potatoes?"



x16 trials,
order randomized

Results: human data

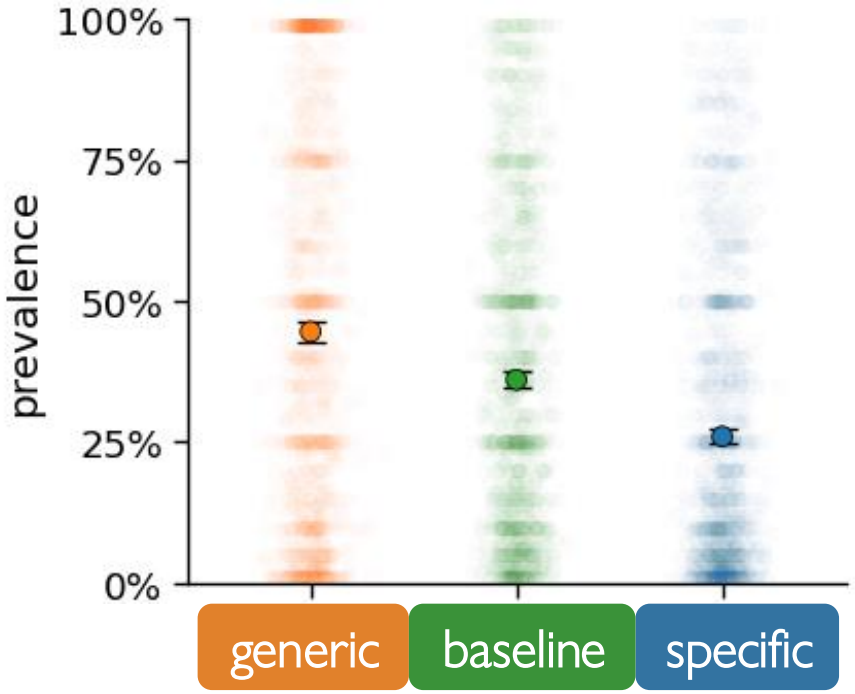


large dots with bars = means with 95% Cis
small dots = individual responses to each trial

Results: human data

As predicted by our model,

1. Hearing generic statements about a category increased generalization across the category.
2. Hearing specific statements about individuals decreased generalization across the category.



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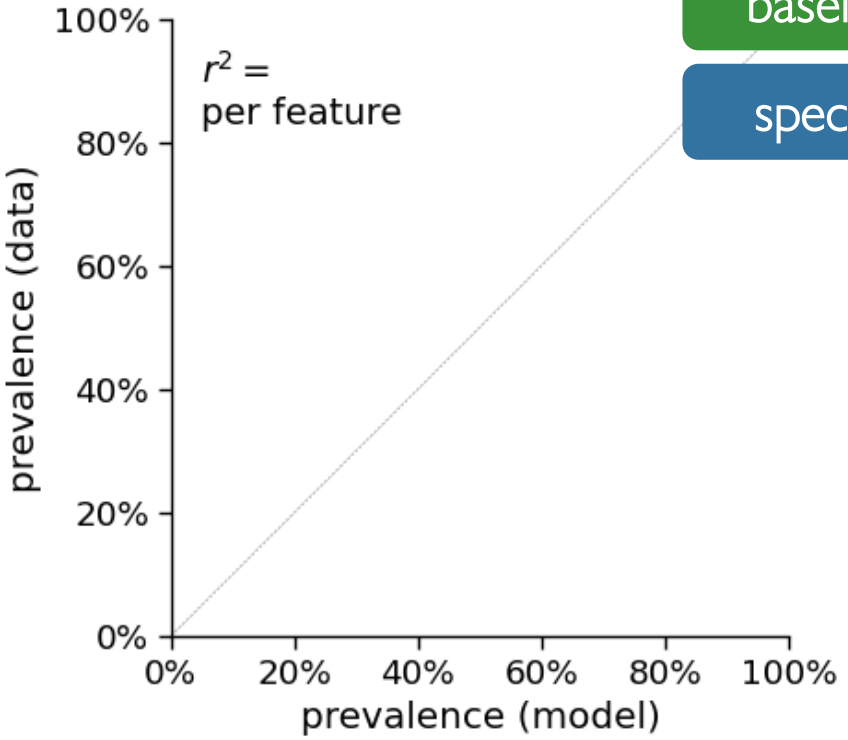
Results: model vs data

conditions

generic

baseline

specific



null model

only reasoning about individual features, ignoring language and pragmatics

n = 284, vs 300 simulated responses from a linking function fit to each feature
(mixture of betas, mixture of category-linked and not-category-linked prevalence weighted by coherence)
dots = mean per feature within condition; large dots = mean per condition

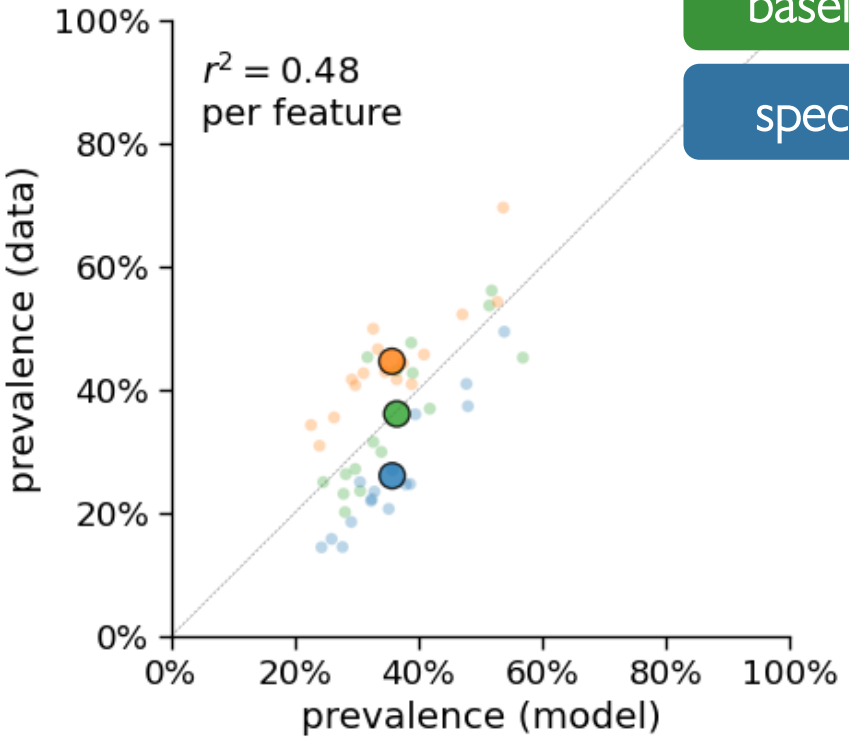
Results: model vs data

conditions

generic

baseline

specific



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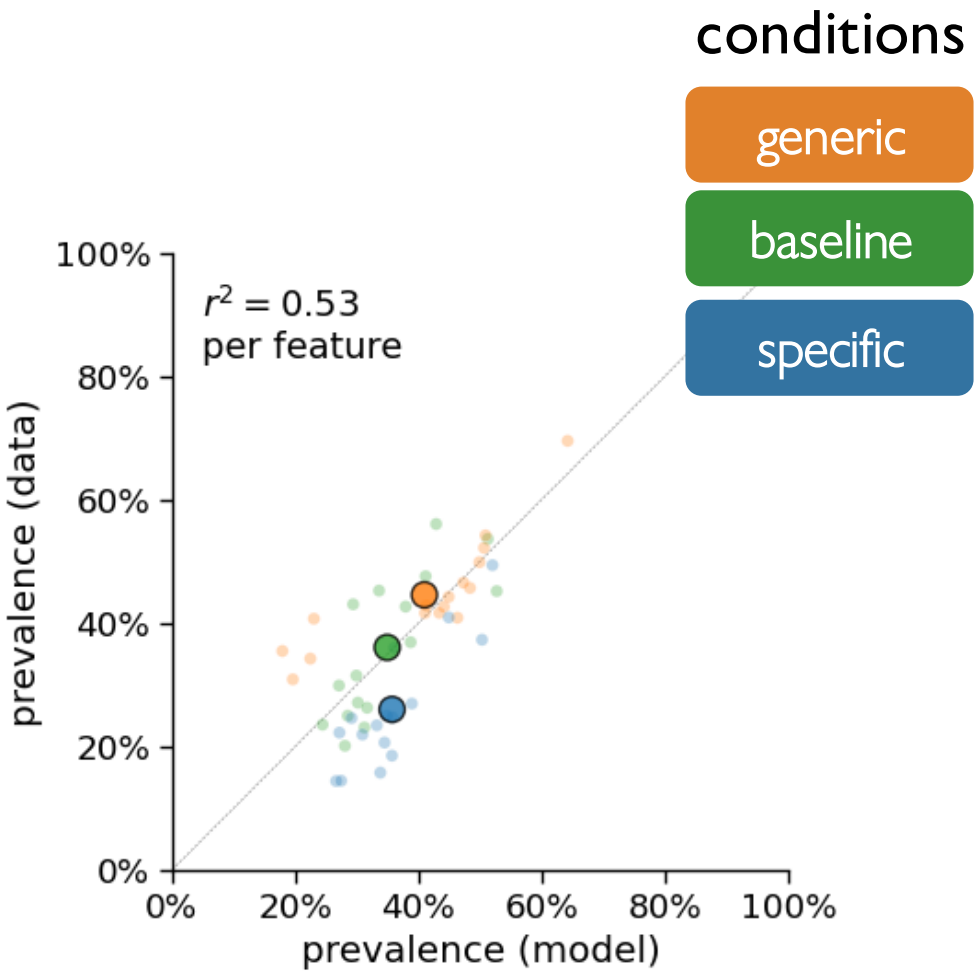
Results: model vs data

Hearing generic statements about a category increases generalization across the category.

→ due to **abstract reasoning** about the literal meaning of language



literal listener model

reasoning about the literal meaning of statements

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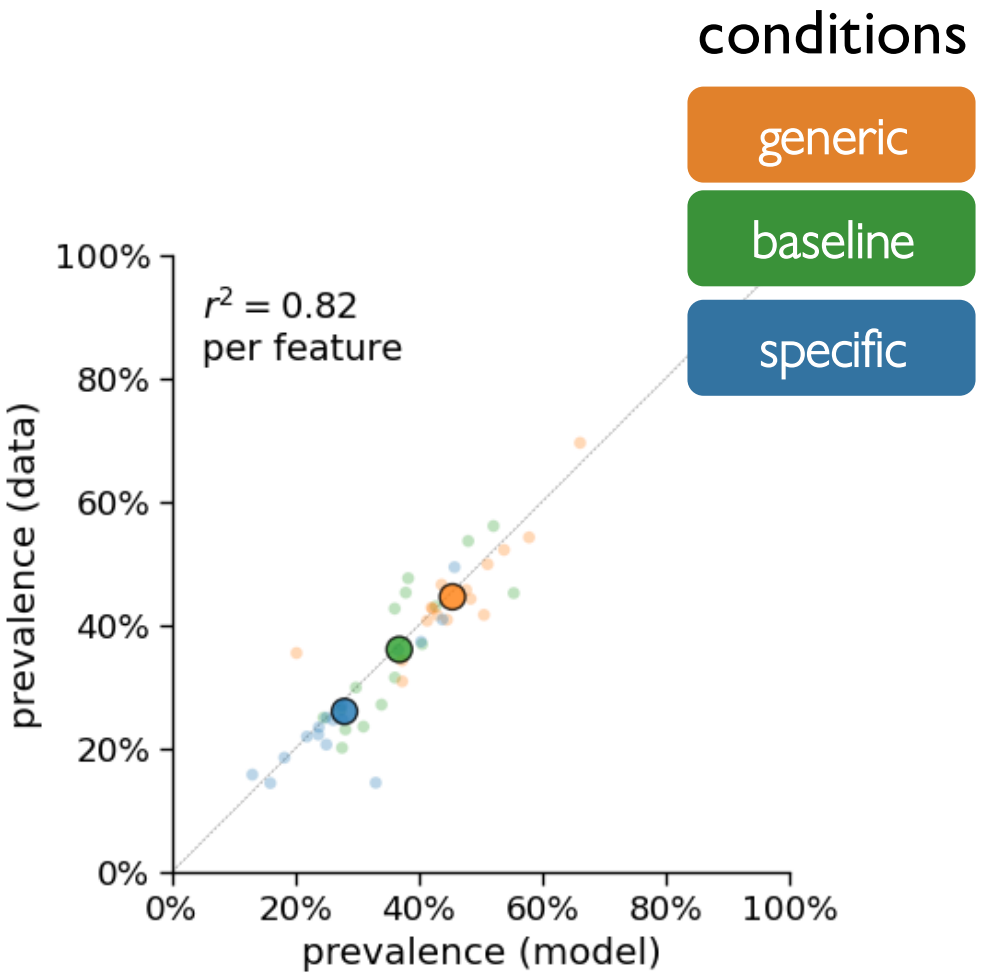
Results: model vs data

Hearing specific statements about individuals decreases generalization across the category.

→ due to **pragmatic reasoning** about what the speaker chose to say



pragmatic listener model

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Language can serve as a signal of the inductive potential of social categories.

Language as a signal of inductive potential

how well a category
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low ⟷ high

categories seen as minimal,
not supportive of generalization

categories seen as coherent/meaningful,
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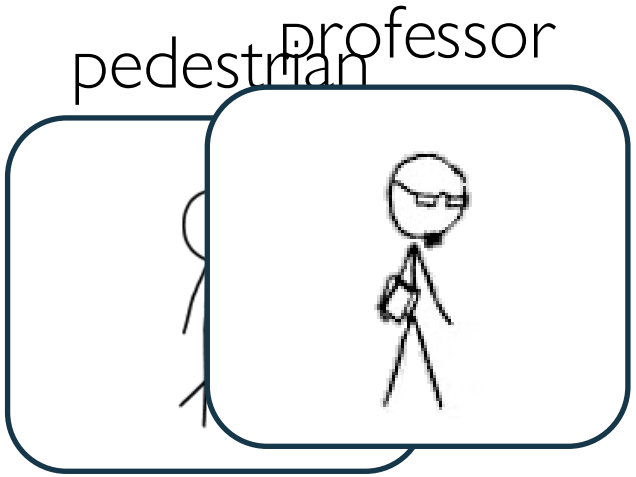

Language as a signal of inductive potential

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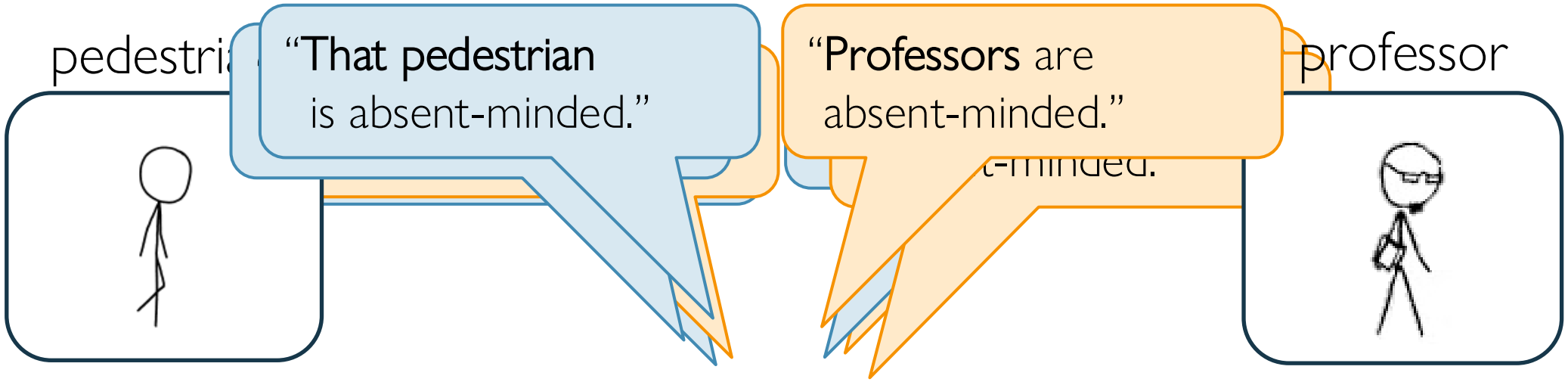
more specifics

more generics



Future directions

How can an understanding of the **relationships between categories** guide generalization?

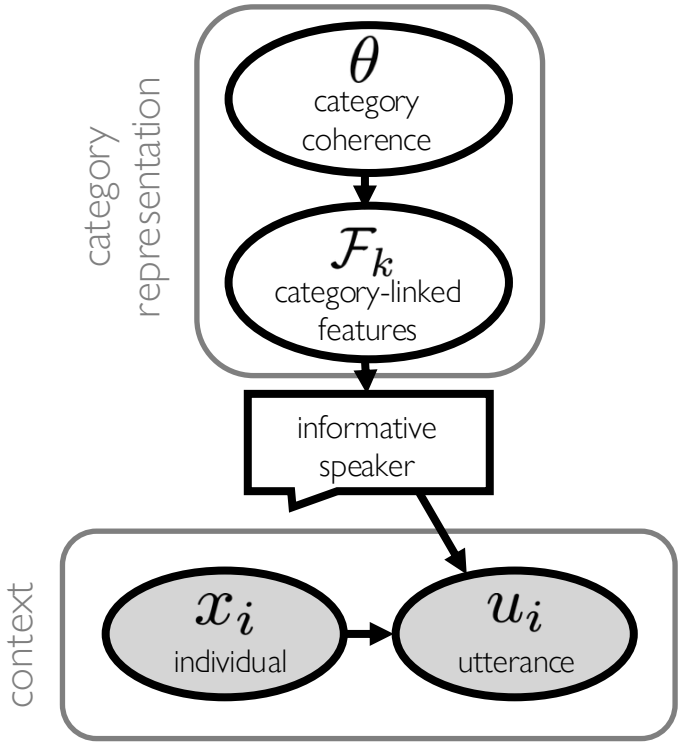
→ over-overhypotheses about e.g., occupational groups

How can an understanding of the **relationships between features** guide generalization?

→ coherence as variable over feature space

If individuals can belong to **multiple categories**, how do we reason about the speaker's choice of which to mention?

→ enriched alternative utterance space



Thank you!

github.com/mariannazhang/compgenerics

labs

Marjorie Rhodes

Mark Ho

CDSC

Computation and Decision-Making Lab

project support

Jess Stephenson
Julia Johnson
participants on Prolific

funding

NYU Postdoctoral Trailblazer Award
NSF SBE Postdoctoral Research Fellowship*
(*until its termination under the present administration)

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Learning about the inductive potential of categories from language

A hierarchical Bayesian + rational speech acts model explains why:

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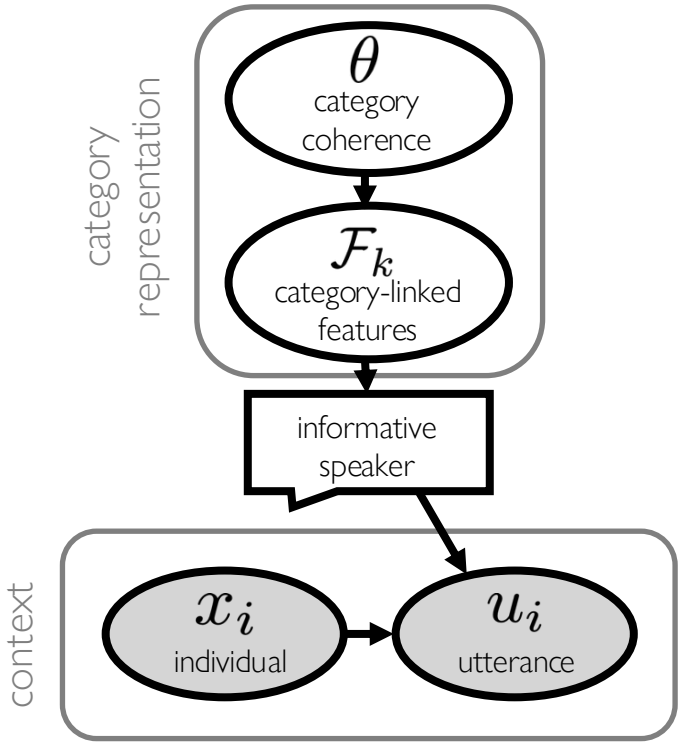
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